DRAFT APPENDIX FOR *AN ORDER ON THE LEECH LATTICE FROM A $\mathbb{Z}_3$-SYMMETRIC TRIPLE-OCTONION PRODUCT* HISTORICAL NOTE: INTEGRAL CAYLEY NUMBERS, 1923–1946

CLAUDE OPUS 4.7

Historical note: integral Cayley numbers, 1923–1946

The construction this paper builds on – an integral-octonion maximal order supported on the E_8 lattice, repaired by a transposition of two imaginary basis units – was developed across four primary papers between 1923 and 1946. Each was obtained in the original and re-derived in exact arithmetic in this project (`python_project/src/verify_{dickson_1923,kirmse_1924,mahler_1942,coxeter_1946}.py`; verification notes in `paper/reviews/`).

Dickson (1923) [Dickson1923] gave the first construction: the Cayley–Dickson doubled product and an explicit 8-element basis of an order O_D, with all 64 basis products closing. His Theorem XV states three maximal orders; the true count is *seven*. The cause is in his derivation (p. 321): he assumes the orders contain the Hurwitz quaternions $\frac{1}{2}(\pm 1 \pm i \pm j \pm k)$, and exactly three of the seven do.

Kirmse (1924) [Kirmse1924] took up the problem independently. His multiplication table is itself a correct octonion algebra. He stated eight maximal orders and exhibited one, J_1 (p. 70); J_1 is not closed, $\frac{1}{2}(1+i_1+i_2+i_3) \cdot \frac{1}{2}(1+i_1+i_4+i_5) = \frac{1}{2}(i_1+i_2+i_4+i_7) \notin J_1$. The true count is again seven.

Mahler (1942) [Mahler1942] developed the arithmetic of a maximal order: for every Cayley number X there is an integral G with $N(X-G) \leq \frac{1}{2}$ (sharper than Dickson's 5/4), and every left or right ideal is principal, generated by an integral Cayley number of norm 1, 2, or 4.

Coxeter (1946) [Coxeter1946] identifies Kirmse's error with the same non-closure witness and records that the remedy is due to R. H. Bruck: "Bruck's domain J can be derived from J_1 by transposing two of the i's" (§4). This is a transposition of two imaginary basis units, a linear involution of $\mathbb{R}^8$; the $\binom{7}{2} = 21$ such transpositions yield the seven maximal orders in seven sets of three. His §§6–13 develop the $E_8/5_{21}$ geometry of the corrected order. A postscript (§14) records his after-the-fact awareness of Dickson and Mahler and identifies Dickson's undercount with exactly the cause above.

The count converges $3 \to 8 \to 7$. The *Coxeter–Bruck transposition* – a transposition of two imaginary coordinate axes – is the historical precedent for the present paper's σ. The two are the same kind of map, a linear involution of $\mathbb{R}^8$, and differ only in what they are asked about: the Coxeter–Bruck transposition acts on a maximal-order candidate; σ acts on the sublattice Ls of Wilson's Leech construction.

References

[Dickson1923] L. E. Dickson, *A new simple theory of hypercomplex integers*, J. Math. Pures Appl. (9) **2** (1923), 281–326.

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Drafted by the AI agent Claude Opus 4.7 (Anthropic) at the direction of Jens Köplinger, as part of the systematic investigation recorded in this repository (`prompt_logs/`). The version that appears in the main paper is owned by the principal investigator.

[Kirmse1924] J. Kirmse, *Über die Darstellbarkeit natürlicher ganzer Zahlen als Summen von acht Quadraten . . .*, Ber. Verh. Sächs. Akad. Wiss. Leipzig, Math.-Phys. Kl. **76** (1924), 63–82.
[Mahler1942] K. Mahler, *On ideals in the Cayley–Dickson algebra*, Proc. Roy. Irish Acad. Sect. A **48** (1942), 123–133.
[Coxeter1946] H. S. M. Coxeter, *Integral Cayley numbers*, Duke Math. J. **13** (1946), 561–578.

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